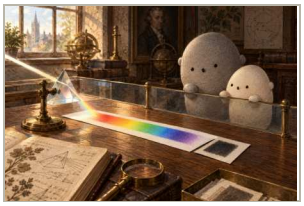
# A little strip of light

The light we cannot see. · Curiosity day

Name _____ Date _____



A simplified reconstruction of Ritter's discovery, not a home experiment. The dark sample is a clue to unseen UV.

A SIMPLIFIED STRIP · POSITIONS ONLY

| Light | UV-A | Violet | Blue | Green | Yellow | Orange | Red |
|---|---|---|---|---|---|---|---|
| **Place** | 1 | 2 | 3 | 4 | 5 | 6 | 7 |

UV-A is unseen. The other six names are visible colors. Real colors blend; these boxes are not equal measured bands.

**1.** There are 7 places. One is unseen UV-A. How many places name visible colors?

$7 - 1 =$ _____ Show your thinking.

**2.** Start at place 1, UV-A. How many one-place hops reach place 7, red?

$7 - 1 =$ _____ Show your thinking.

**3.** Green is at place 4. Red is at place 7. How many places farther along is red?

$7 - 4 =$ _____ Show your thinking.

Wavelength is the space from one wave crest to the next. Imagine how stretched out a drawn wave looks. Along our strip toward red, wavelengths get longer. Which end would have more closely spaced crests?



WONDER DAILY

# The light we cannot see.

## Make a little book



**1** Fold it into 8 boxes. Open it flat.



**2** Fold short end to short end. Cut the thick line. Open it.



**3** Fold the long way, pictures out. Push the ends in.




**4** Fold into a book. Page 1 goes on the front!





## Keep the clue

The museum lamp switches off. The green fluorescence stops almost at once. Little sketches a lamp, a star, and light coming out.





## Light in, light out

The green star has a special coating: a fluorescent phosphor. It absorbs UV-A and quickly emits green light.





## Two paper stars

A green star shines in a quiet museum case. Beside it, a cream star stays dull. Both sit under the same dark-looking lamp.





## Beyond violet

Their search begins long ago. In 1801, Johann Ritter spread sunlight into a rainbow.





## A name for the glow

Another display tells of George Stokes. In 1852, he studied materials that took in light and gave out a different color.





## A dark-looking light

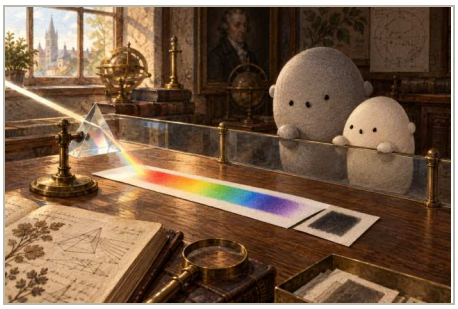
Back at the stars, Little studies the black light. It gives out mostly UV-A, the ultraviolet near visible violet. Our eyes cannot see UV-A.

# Give the glow a path.

The light we cannot see. · Curiosity day

Name _____ Date _____

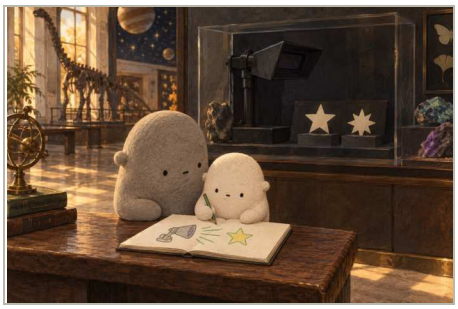
Choose one scene. Say what the clue helped Little discover. Draw, label, or write one idea.



Beyond violet



Light in, light out



Keep the clue

Draw a clue from your scene. Add an arrow to show where light goes or a label to name a change.

UV-A · phosphor · absorb · emit · fluorescence · visible · clue

The dark patch was a clue because … What was reaching it?

The coating takes in … What light goes in? What light comes out?

When the lamp goes off, … What happens to fluorescence?

Point to the detail your words explain. If you write a sentence, check its first capital and ending mark.